

AMD vLLM AMD Instinct MI300X ● Live

ShelfWise

Agentic AI store operations, grounded in real math.

An AI system that runs a supermarket's daily decisions — expiry markdowns, procurement, pricing integrity, cold chain — through real tool-calling agents on AMD compute, with a human approving every action.

THE PROBLEM

A supermarket makes hundreds of interlocking decisions a day. Most are made late, or on gut feel.

Expiry & waste

Which stock to mark down **before** it expires — and by how much — is a forecasting and simulation problem, not a guess.

Procurement & sourcing

When a shortage hits, "transfer some stock" isn't an answer. From **which branch? The DC? A supplier?** At what lead time?

Pricing integrity

Till prices drift from catalogue prices. Every unnoticed mismatch is silent margin loss or a compliance risk.

Cold chain

A fridge outage starts a countdown on thousands of rand of stock. Risk must be **scored**, not eyeballed.

Every answer flows through real agents calling real tools on AMD compute.

1 · Events in

POS sales, stock updates, scans, sensor alerts and connector webhooks land on one event bus.



2 · Agents reason

Critic / Executive agent pairs — **Gemma-4-E4B-it on AMD Instinct MI300X via vLLM** — run a bounded tool-calling loop.



3 · Tools compute

11 read-only tools: stock, demand forecast, expiry risk, reorder policy, supplier ranking, sourcing, cold-chain risk, price integrity, markdown simulation, decisions, thresholds.



4 · Human approves

Every recommendation lands as a pending decision. Nothing writes back without sign-off. Full audit trail.

The grounding guarantee: a final answer must cite the numbers its tools actually computed. An answer citing a number no tool produced is **rejected and re-run** — **never shipped**. The model uses tools as its calculator; it cannot invent figures.

Chat is the front door — agentic, structured, honest.

- One question can trigger **4+ live tool calls** in a single turn — approvals, stock, deliveries, sourcing.
- Answers render as **structured reports**: headings, bullets, bolded figures — readable by a store manager, not a data dump.
- When it lacks data, it **says so** — it never fabricates a reading.
- Multi-user, multi-conversation, tenant-isolated. The authenticated tenant always overrides anything the model writes into tool arguments.

We are short on SKU 4011. Where should the replacement stock come from?

- **ShelfWise**

I recommend taking **6 units** from branch **store_02_sandton** — it is the closest at **5.00 km** with the fastest delivery time of **2.00 hours**, beating store_09_midrand (4.00 h away).

This covers only part of the shortage: place a **purchase order for the remaining 12 units**.

Real answer from the live MI300X endpoint. It ranked branches, the DC, and suppliers by availability, distance, and lead time — and explained why the winner won.

PROOF, NOT PROMISES

Stress-tested live before anything was demoed.

A 15-minute continuous world simulation drove the full public API against the live AMD MI300X endpoint — every scenario type, every cascade, connectors, approvals, learning — with hard failure on any offline fallback.

333/333

chat calls genuinely model-backed
zero offline fallbacks, **zero** errors

4,618

decisions created
zero ID collisions

2,990

human approve / reject cycles
zero mismatches

418

automated tests passing

< 6 s

multi-tool report answer,
end to end

165

machine-verified capabilities in a committed
manifest — CI fails on drift

AMD compute is the engine, not a checkbox.

Inference

- **AMD Instinct MI300X** (AMD Developer Cloud droplet)
- **vLLM 0.23 on ROCm**, serving **google/gemma-4-E4B-it**
- Native tool calling with the Gemma-4 parser — the agent loop runs on the GPU, not around it
- 131k-token context window powering long, tool-heavy agent conversations

Application

- **FastAPI** backend — event bus, cascades, HITL, learning loop, write-back queue
- **React + Vite** chat-first console with live approval queue
- Custom decision-science layer: reorder policy, demand forecasting, expiry & cold-chain risk scoring, markdown simulation, multi-source sourcing optimisation
- All math in tested Python tools — **never hidden inside prompts**

Honest scope: demo data is a seeded single-store scenario. The decision logic is general and unit-tested independently of the data; wiring live multi-branch feeds is roadmap, and we say so.

WHERE THIS GOES

Every store position, one governed AI colleague.

Next

Live connector feeds (POS / WMS / ERP), multi-branch inventory, PDF stock sheets and period reports generated from the same grounded tools.

Why it wins

Agents that must show their math, a human on every action, and receipts for every claim — governance is the feature, not the afterthought.

Why AMD

One MI300X proved the full sequential live workload with sub-6-second multi-tool answers. Concurrent capacity benchmarking is the next evidence gate.

■ **ShelfWise** — agentic reasoning, grounded in real tools, governed by a human, running on AMD.